SURAWIWAT SCHOOL

Real-Time Hand Tremor Suppression System Utilizing Deep Neural Network on Edge Computing

Presented by

Mr. Takorn Wisetrith
Mr. Lertpisit Puengwongsanurak
Mr. Waritnan Waewbunditwittaya

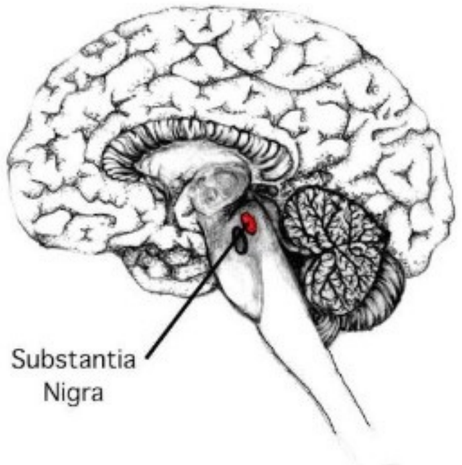
Project Advisor

Prof. Aphilak Lonklang
T. Nongnud Paiboon

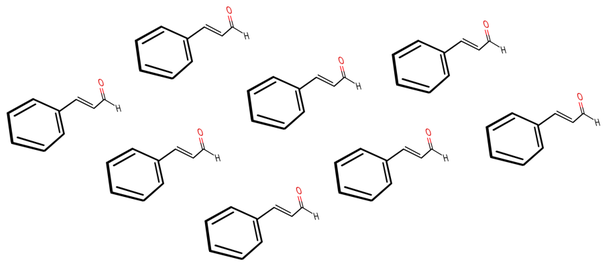


Introduction

Normal Person



produce

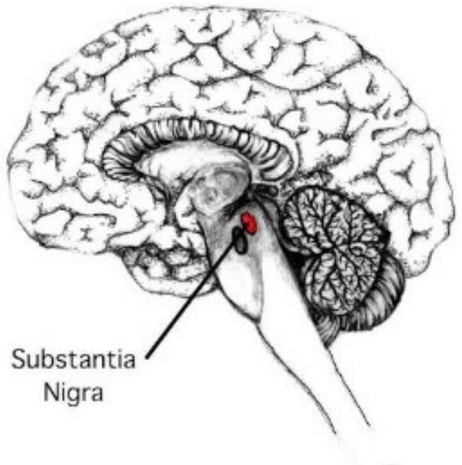


Sufficient dopamine

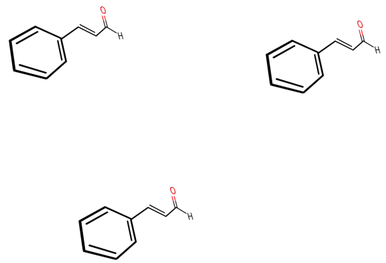


Introduction

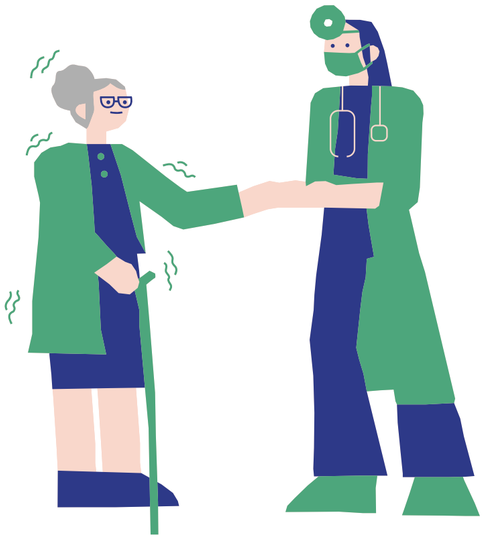
Parkinson Disease



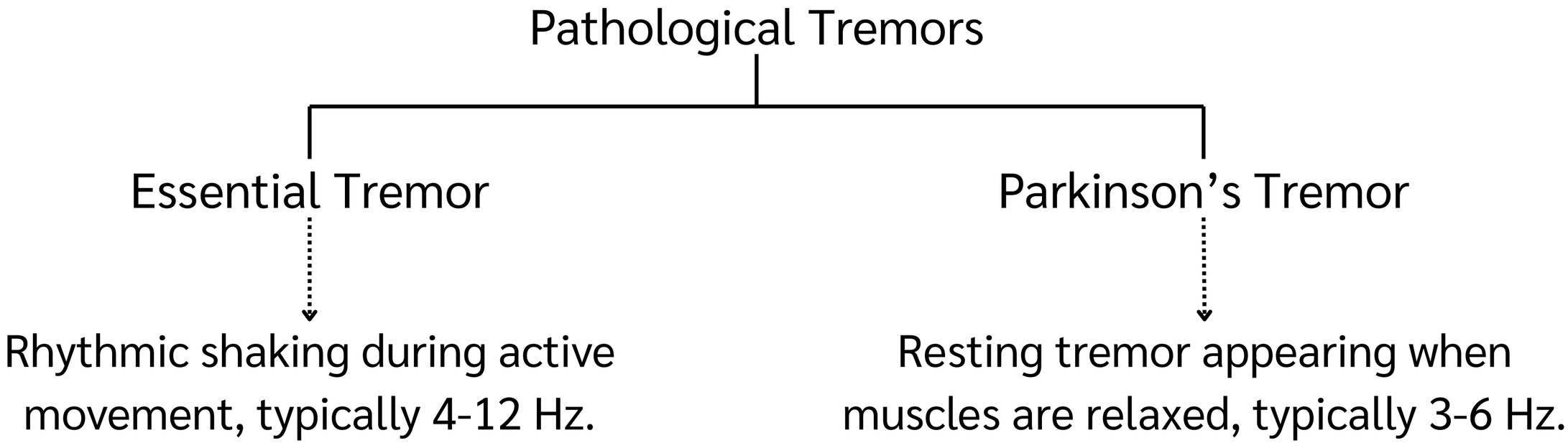
produce



Insufficient dopamine

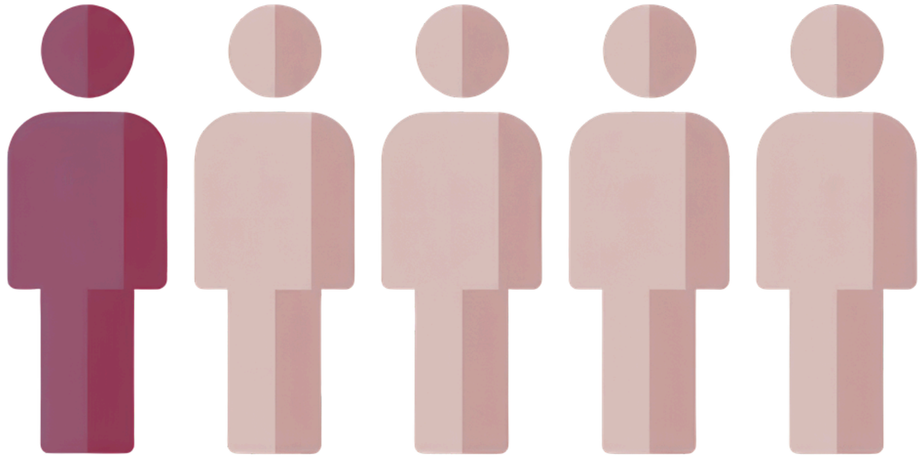


Statement of the problem



Statement of the problem

Over 70% of patients experience challenges in living independently due to uncontrollable hand tremor



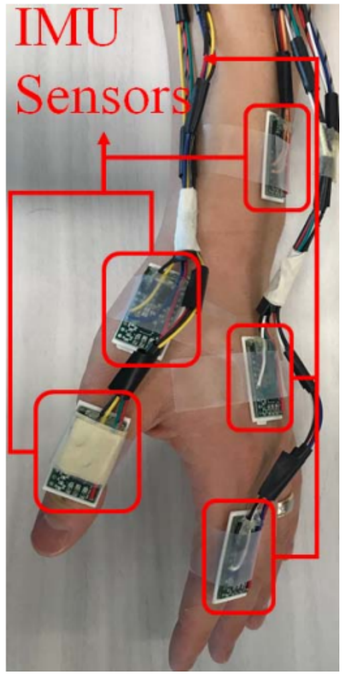
Objectives

- To enhance the overall quality of life and mental well-being of elderly users.
- To improve daily living capabilities by mitigating physical barriers caused by hand tremors.
- To evaluate the wearable device's efficacy in reducing active tremors and optimize its design for individual ergonomic needs.



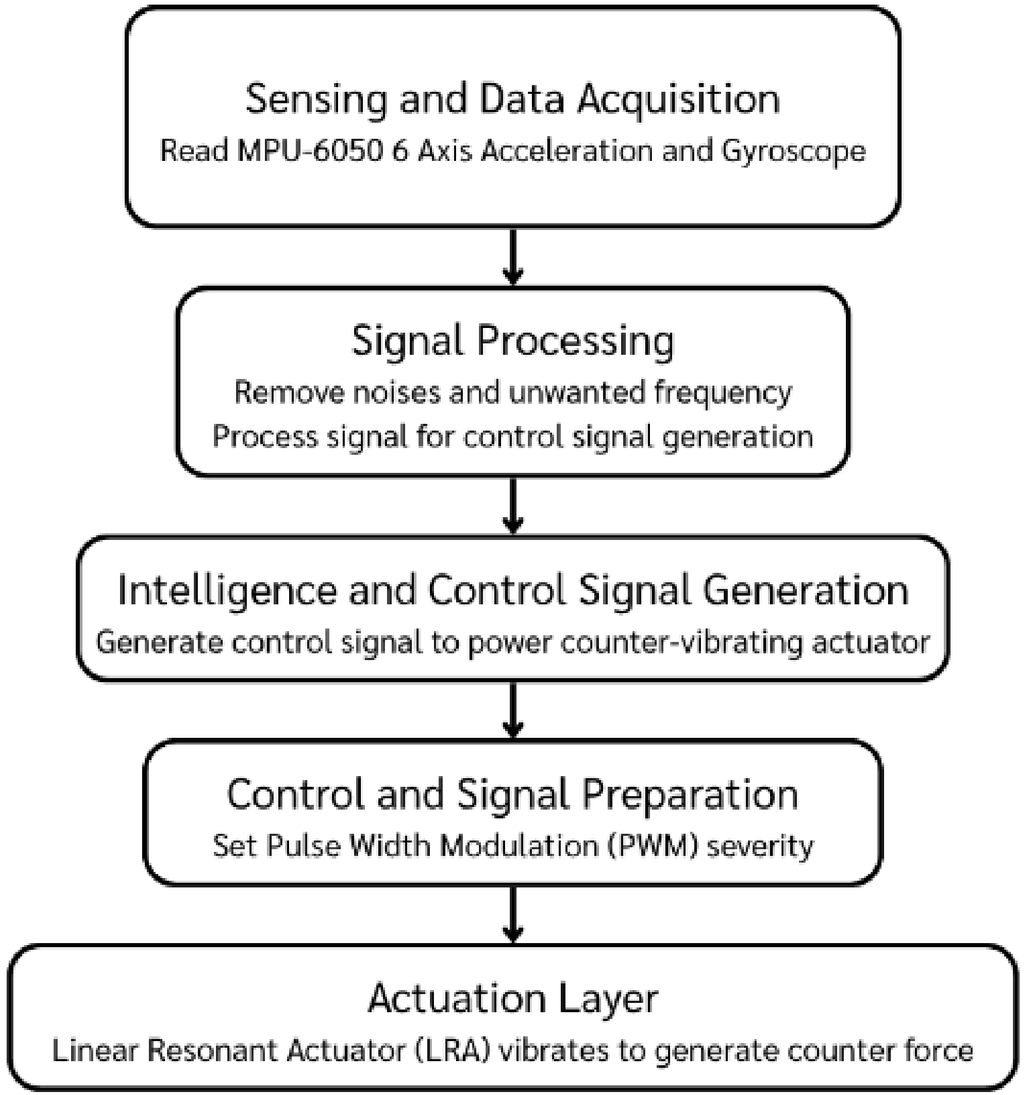
Literature review

Ibrahim, A., Zhou, Y., Jenkins, M. E. Real-Time Voluntary Motion Prediction and Parkinson's Tremor Reduction Using Deep Neural Networks. IEEE Transactions on Neural Systems and Rehabilitation Engineering 2021, 29(1), 1413-1423.



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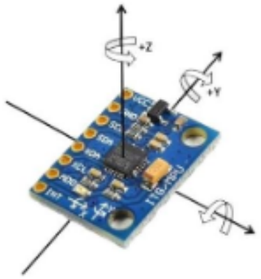
System Architecture and Design

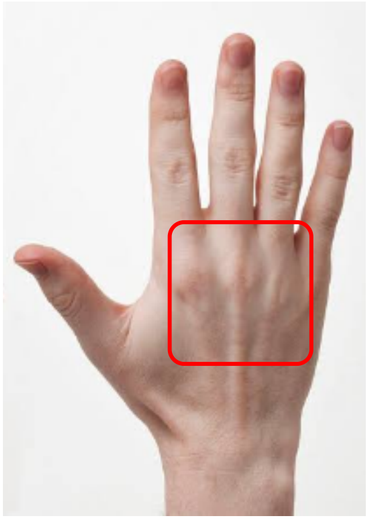


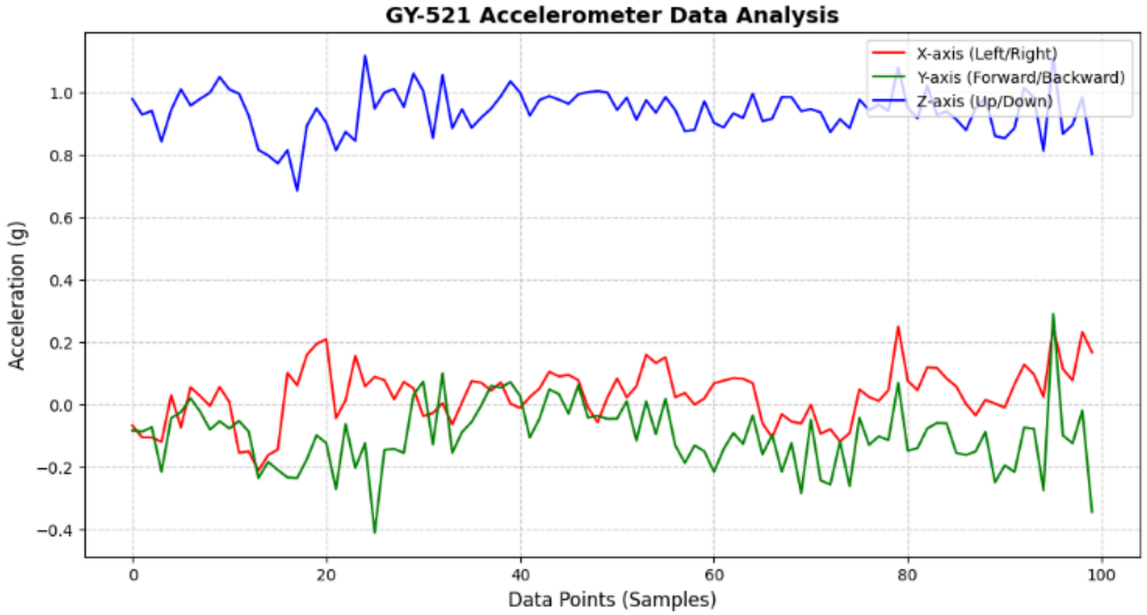
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System Architecture and Design

1. Sensing Layer : MPU6050 (GY-521)



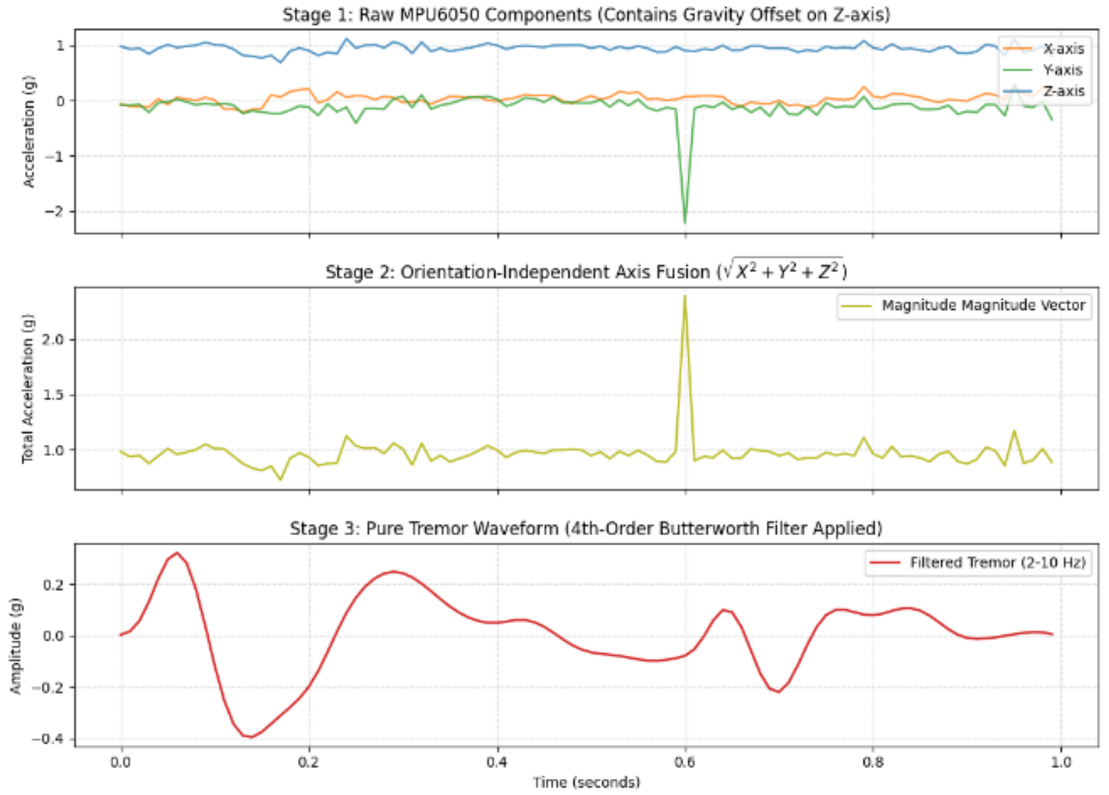




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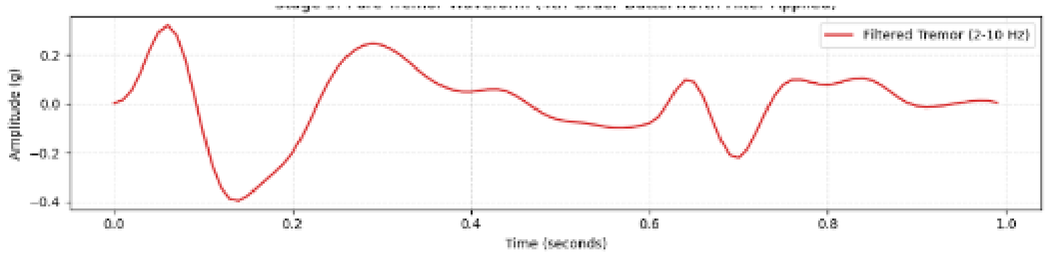
System Architecture and Design

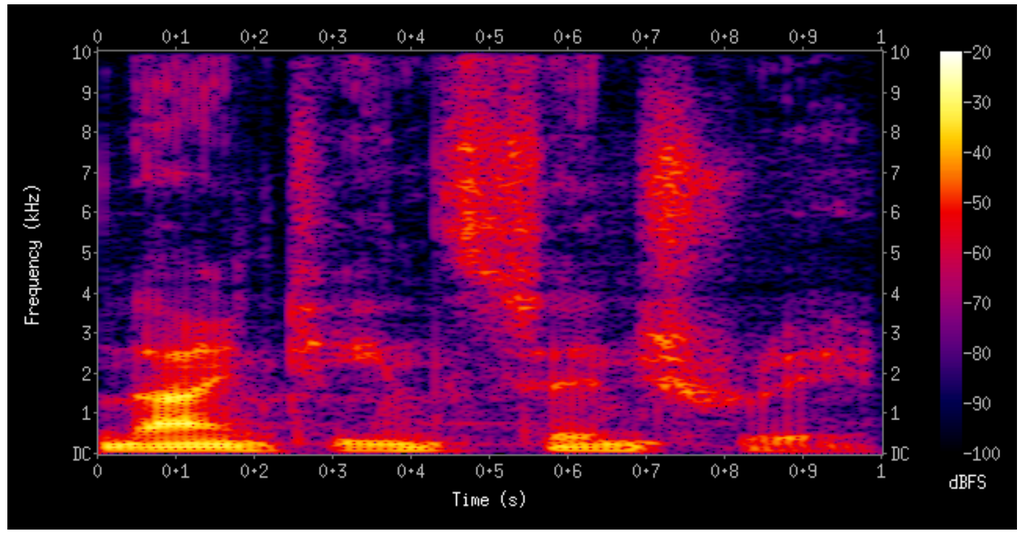
2. Signal Processing : Butterworth Bandpass Filter



System Architecture and Design

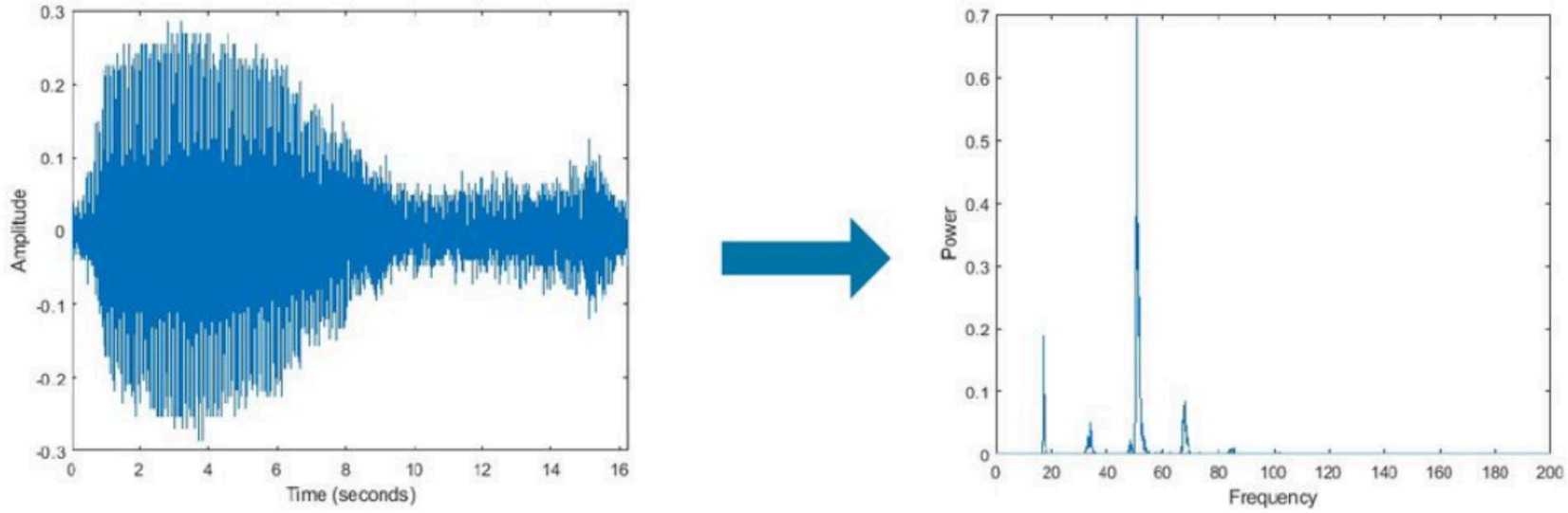
2. Signal Processing : Short-Time Fourier Transform (STFT)





System Architecture and Design

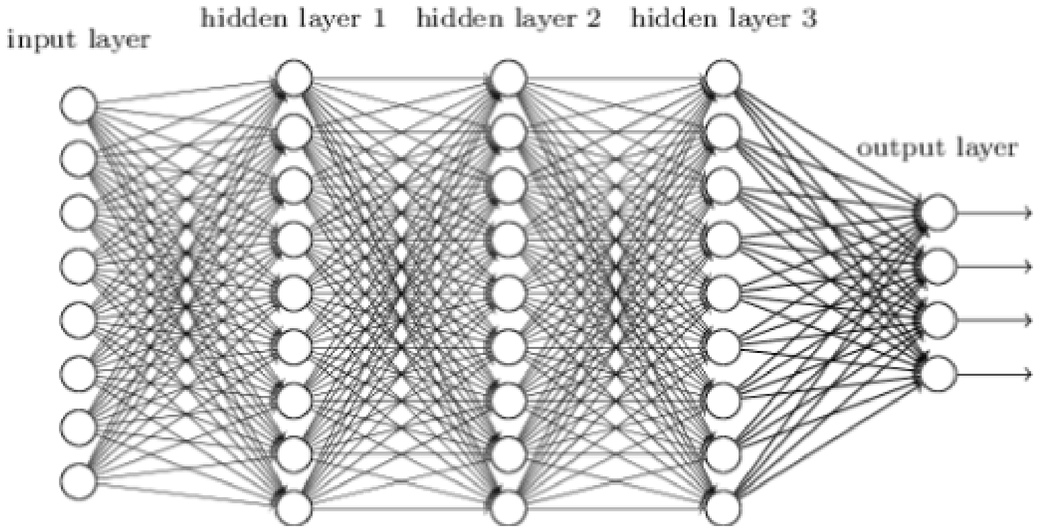
2. Signal Processing : Short-Time Fourier Transform (STFT)



System Architecture and Design

3. Intelligence : Deep Neural Network (DNN)

- Root Mean Square
- Peak Frequency
- Spectral Power
- Spectral Entropy

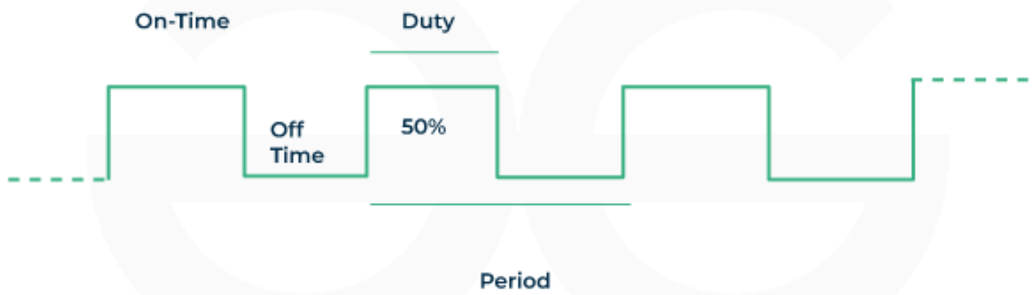


- Actuator Motor Speed
- Suppression Frequency

System Architecture and Design

4. Control Layer : DRV2605L Haptic Driver Module





System Architecture and Design

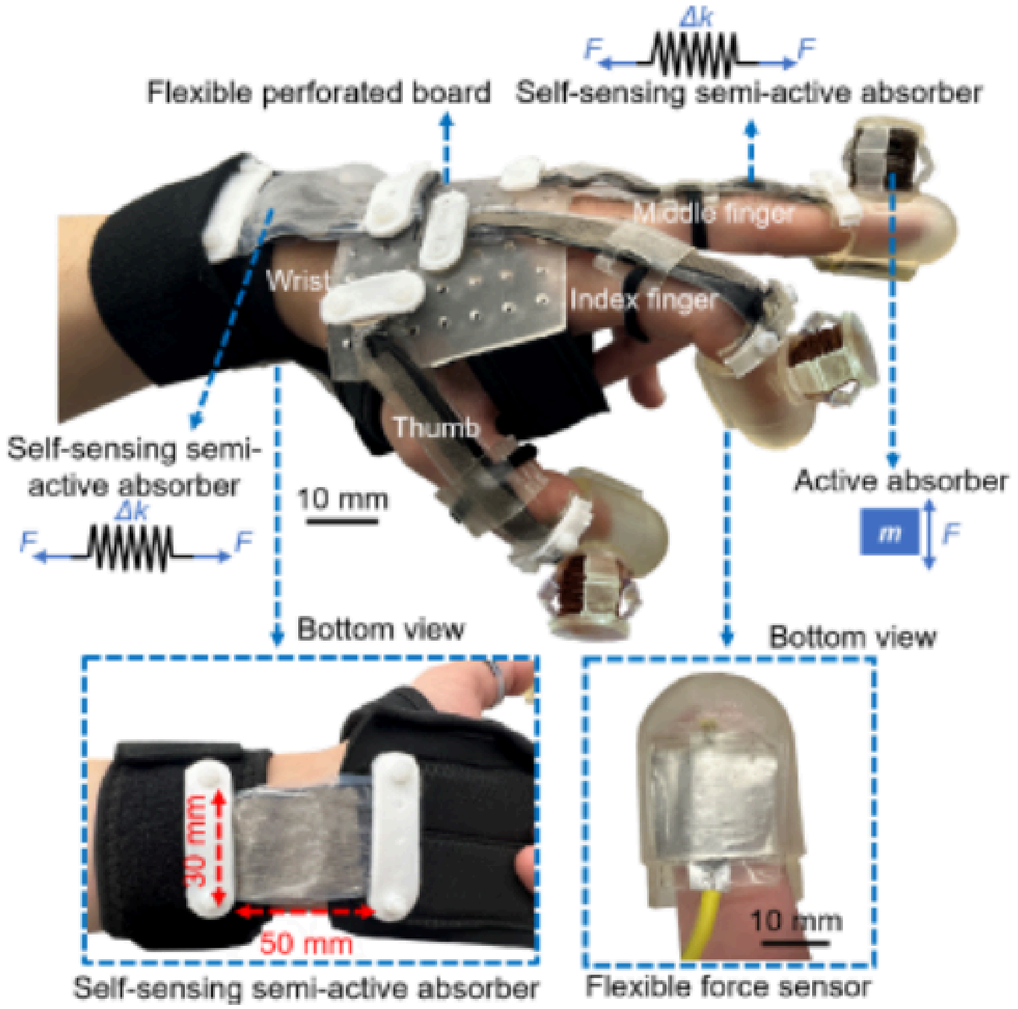
- Fast Respond
- Low Power Consumption
- Can Finely Adjust Frequency And Amplitude



Prototype

Smart Glove :Focusing on aesthetics with a lightweight and discreet design.

To reduce the clinical aesthetic, the device features a fitness-glove design that enhances user confidence during public use.

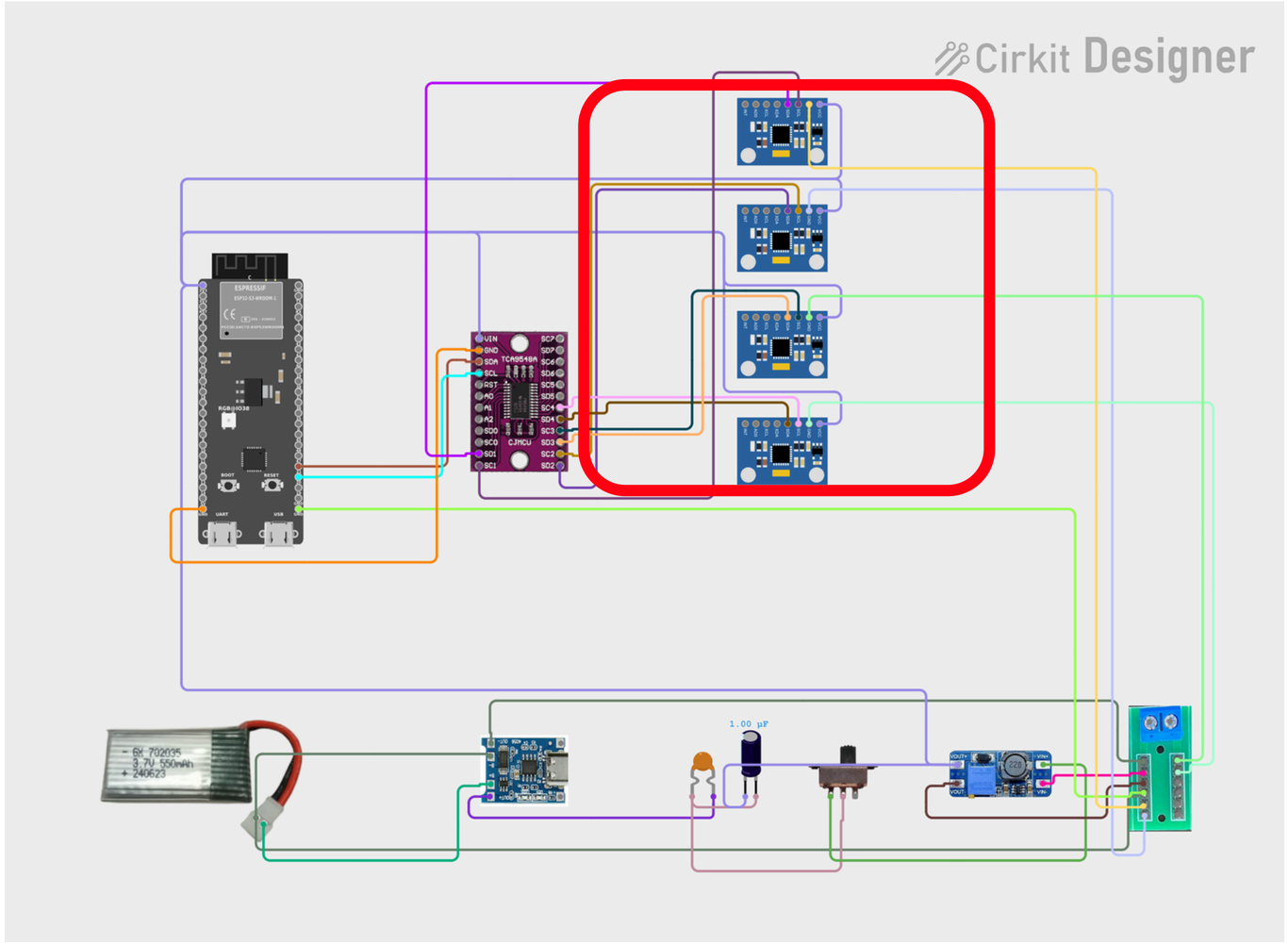


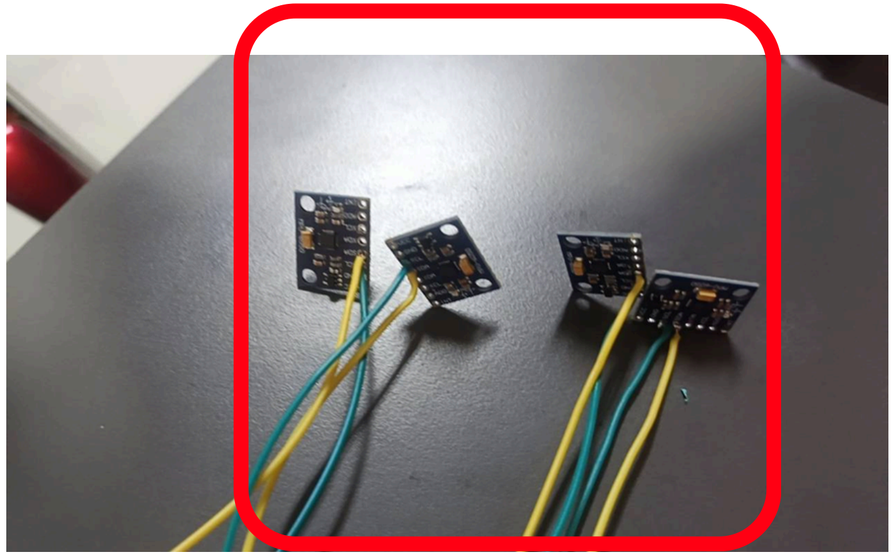
Tecnical information

Total weight: 320 grams
Materials:Neoprene fabric and silicone
Battery life: 6–8 hours per charge

current stage

Architecture design





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Plans

Plan	Jun 69	Jul 69	Aug 69	Sep 69	Oct 69	Nov 69	Dec 69	Jan 70	ผู้รับผิดชอบ
Search for problems of interest									
Literature review									
Ethics Training									
Glove Design									
Experimental Design									
Ethical Approval Application									
Data-Collecting									
Reporting of Results									

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11.Supto, S. S., Abir, S. M., Islam, M. A. Smart Hand Tremor Suppression and Rehabilitation Glove for Parkinson's Patients. Senior Design Project Report, Department of Electrical and Computer Engineering, North South University 2025, 1(1), 1-35.
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Thank you for your attention